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Decoupled information-guided cross-modal alignment for visible-infrared person re-identification

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ABSTRACT

Visible–infrared person re-identification (VI-ReID) is an important technique for around-the-clock person matching, and its primary challenge arises from substantial cross-modal discrepancies. To address this challenge, we propose a Decoupled Information-Guided Cross-Modal Alignment (DIGCA) framework organized into three stages: multi-scale feature modeling, partial functional decoupling, and guided cross-modal alignment. First, a Hierarchical Context Extractor (HCE) aggregates multi-scale contextual information through dilated convolutions and residual connections to enrich the initial identity representation. Second, a Multi-Branch Unified Encoder (MBUE) organizes complementary feature streams through parallel global-relation and local-spatial modeling. The resulting representations exhibit different information tendencies and promote partial functional decoupling between identity-related and modality-related information. Finally, the Decoupled Information-Guided Cross-Modal Alignment (DIG-CMA) module refines the two streams with channel and spatial attention and integrates them through cross-attention. Under the joint optimization objective, the resulting representations support cross-modal alignment. Experiments on SYSU-MM01, RegDB, and LLCM show that DIGCA achieves competitive overall performance compared with recent VI-ReID methods, providing empirical support for the effectiveness of the proposed decoupling-guided alignment strategy in cross-modal identity matching.

1. Introduction

Person re-identification (ReID) aims to identify and match a target person across different cameras or across different time periods of the same camera. As an important enabling technology for public security and intelligent surveillance, ReID has considerable research and practical value (Wu et al., 2024). Most existing ReID methods are designed for visible-light imagery, and their performance can degrade substantially in poorly illuminated or nighttime environments, limiting their applicability to around-the-clock intelligent surveillance. Visible–infrared person re-identification (VI-ReID) has therefore emerged to address this limitation. VI-ReID uses visible (VIS) cameras to capture pedestrian images during the daytime and infrared (IR) cameras to capture images at night, thereby enabling cross-modal person retrieval (Zhang & Wang, 2023) and providing a practical solution for around-the-clock surveillance.

Compared with conventional single-modal ReID, VI-ReID faces additional challenges (Liang et al., 2023). As shown in Fig. 1, conventional ReID already needs to address intraclass variations and interclass similarities caused by viewpoint changes and similar pedestrian appearances (see Fig. 1(a) and (b)). Infrared images, however, primarily reflect pedestrian contours and thermal distributions and differ inherently from visible images in visual characteristics such as texture and color. VI-ReID must therefore address not only the inherent challenges of conventional ReID but also the substantial discrepancies between the visible and infrared modalities (see Fig. 1(c)) (Wei et al., 2024). These discrepancies increase the difficulty of cross-modal feature matching and limit recognition accuracy.

To reduce cross-modal discrepancies, recent VI-ReID research has progressed along three interrelated directions. The first direction learns modality-invariant representations through shared feature spaces, feature disentanglement, or modality perturbation, with the aim of reducing the influence of modality-related factors on identity matching (Ren & Zhang, 2024; Tao et al., 2025; Ye et al., 2024; Zuo et al., 2025). The second direction aligns visible and infrared representations through image translation, intermediate-modality construction, and metric or distribution constraints (Dai et al., 2025; Liu et al., 2023b; Wang et al., 2019a, 2020, 2019b; Zhang et al., 2021, 2025). The third direction employs attention mechanisms, Transformers, or multi-granularity modeling to integrate identity-discriminative cues across body regions, scales, and modalities, thereby enhancing cross-modal feature interaction (Feng et al., 2025a, 2023b; Liang et al., 2023; Lu et al., 2023).

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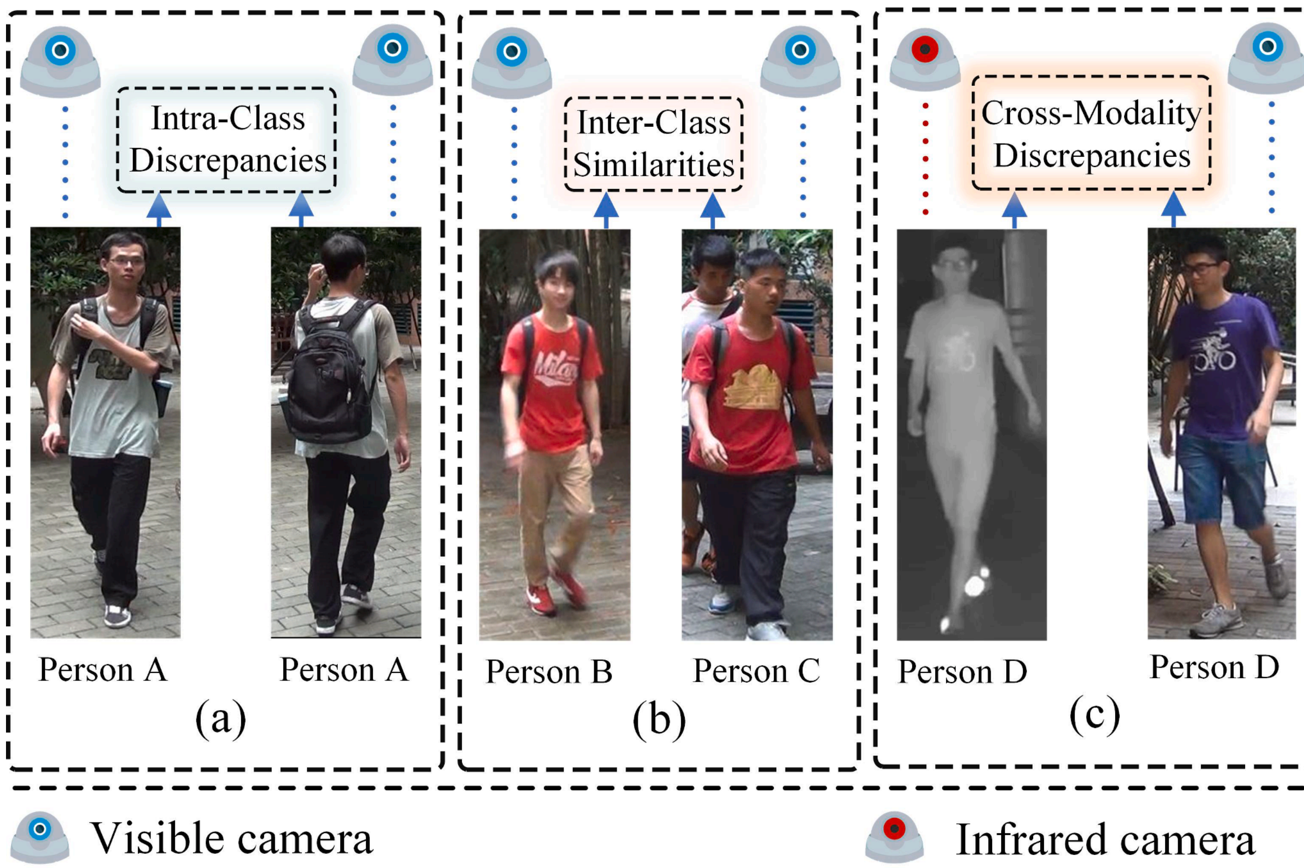


Fig. 1. Intra-class differences (a) and inter-class similarities (b) within a single modality, along with cross-modal discrepancies (c), collectively constrain the performance of the VI-ReID task.

These lines of research have advanced VI-ReID from the perspectives of representation learning, embedding alignment, and feature interaction. However, in many methods, functional feature organization and subsequent cross-modal alignment remain relatively separate design stages. Disentanglement-based methods generally focus on distinguishing modality-shared information from modality-related information, while how the resulting functionally differentiated representations can provide targeted guidance for subsequent alignment remains underexplored. Meanwhile, the effectiveness of alignment and interaction modules depends on their input representations. When identity-related and modality-related information remain entangled, modality bias may affect feature selection and aggregation. Motivated by this observation, we investigate a decoupling-guided alignment strategy that first organizes complementary representations with different information tendencies and then uses them to guide feature-stream interaction for cross-modal alignment.

To instantiate this strategy, we develop the Decoupled Information-Guided Cross-Modal Alignment (DIGCA) framework. DIGCA connects multi-scale feature modeling and functional feature organization with subsequent cross-modal alignment, allowing the organized complementary representations to guide the interaction process rather than treating all extracted cues homogeneously.

Specifically, the Hierarchical Context Extractor (HCE) aggregates multi-scale contextual information through dilated convolutions with different dilation rates and residual connections to enrich the initial identity representation. The Multi-Branch Unified Encoder (MBUE) organizes complementary feature streams through parallel global-relation and local-spatial modeling, thereby promoting partial functional decoupling between identity-related and modality-related information. Based on these functionally differentiated representations, the Decoupled Information-Guided Cross-Modal Alignment (DIG-CMA) module applies channel and spatial attention to refine the global-relation and local-spatial streams, respectively, and integrates the two streams through cross-attention. The resulting representations are jointly optimized to support cross-modal alignment. Experiments on SYSU-MM01, RegDB, and LLCM show that DIGCA achieves competitive performance compared with recent VI-ReID methods, with strong results on SYSU-MM01 and LLCM and comparable results on RegDB.

The main contributions of this work are summarized as follows:

- We formulate a decoupling-guided alignment strategy for VI-ReID. The strategy uses functionally differentiated representations to guide the interaction between complementary feature streams, aiming to alleviate cross-modal discrepancies.
- We develop a Hierarchical Context Extractor (HCE) that aggregates multi-scale contextual information using dilated convolutions with different dilation rates and residual connections, thereby enriching the initial identity representation extracted by the backbone.
- We develop a Multi-Branch Unified Encoder (MBUE) that organizes complementary feature streams through parallel global-relation and local-spatial modeling. The resulting representations exhibit different information tendencies and promote partial functional decoupling between identity-related and modality-related information.
- We design a Decoupled Information-Guided Cross-Modal Alignment (DIG-CMA) module that uses channel and spatial attention to refine the global-relation and local-spatial streams, respectively, and then integrates the two streams through cross-attention to support cross-modal alignment.

The remainder of this paper is organized as follows. Section 2 reviews VI-ReID studies related to this work. Section 3 describes the proposed DIGCA framework and its joint optimization objective. Section 4 presents the experimental settings and results, followed by analyses of overall performance, component contributions, the functionally differentiated behavior of MBUE, training stability, and computational efficiency. Section 5 concludes the paper.

2. Related work

2.1. Feature disentanglement

Visible and infrared images of the same identity share semantic content but differ substantially in spectrum-dependent appearance. Feature disentanglement methods seek to preserve identity-discriminative cues while regulating modality-related variations. A common route decomposes representations into modality-shared and modality-specific components, with adversarial objectives used to suppress modality predictability, reconstruction losses used to retain input information, and orthogonality constraints used to reduce redundancy between components (Wu et al., 2017; Ye et al., 2021). These constraints encourage different functional roles, although they do not by themselves imply strict information separation. Accordingly, the learned components are typically optimized jointly rather than interpreted as independently identifiable sources of identity and modality information.

Recent methods extend this idea through perturbation, knowledge transfer, and multi-level regulation. DARD (Wei et al., 2024) perturbs modality-related characteristics through image-level channel exchange and feature-level magnitude variation, thereby increasing the robustness of the learned shared representation. IDKL (Ren & Zhang, 2024) purifies identity-aware cues contained in modality-specific features and distills them into the shared representation. TMD (Lu et al., 2024) regulates modality information at the image, feature-distribution, and instance-feature levels, whereas cooperative separation jointly learns shared and specific components (Yang et al., 2024). Memory attention and orthogonal decomposition have also been combined to regulate modality and posture factors (Lu et al., 2025). These studies show that the utility of the learned components depends not only on how they are defined and constrained, but also on how they contribute to subsequent matching or alignment.

2.2. Cross-modal alignment

Cross-modal alignment reduces visible–infrared discrepancies at the image, feature-distribution, or embedding-space level. Image-level methods translate or synthesize samples before feature extraction. Joint pixel and feature alignment (Wang et al., 2019a), cross-modality paired-image generation (Wang et al., 2020), and unified middle-modality learning (Zhang et al., 2021) provide progressively more comparable inputs. Related work shifts compensation to the representation space: FMCNet generates missing modality-specific information at the feature level (Zhang et al., 2022), MRCN restores identity-relevant information removed during normalization and compensates features across modalities (Zhang et al., 2023), DEEN expands embeddings to expose diverse cross-modality cues (Zhang & Wang, 2023), and AMML constructs adaptive middle modalities at both image and feature levels (Zhang et al., 2025).

Feature- and metric-level approaches instead constrain the learned space directly. Modality confusion with center aggregation promotes cross-modal compactness (Hao et al., 2021), while channel augmentation perturbs spectral cues to improve robustness (Ye et al., 2021a, 2024). Memory-augmented unidirectional metrics regulate asymmetric cross-modal distances (Liu et al., 2022b); related center, triplet, distribution, and graph constraints further balance intra-class compactness and inter-class separation. Such objectives improve embedding consistency, while their effect also depends on the identity cues and modality information retained in the representations being aligned.

2.3. Attention and multi-granularity modeling

Attention and Transformer architectures support feature selection, interaction, and long-range relation modeling across spatial regions and representation levels. The Cross-Modality Interaction Transformer models dependencies between heterogeneous representations (Feng et al., 2023b), whereas the Cross-Modality Transformer with Modality Mining explicitly exploits modality information to construct discriminative embeddings (Liang et al., 2023). The Progressive Modality-Shared Transformer introduces an auxiliary grayscale modality and progressive learning to strengthen reliable shared cues (Lu et al., 2023). CFET combines Transformer-based extraction with multi-granularity enhancement and channel interaction fusion (Guo et al., 2025a), while CM^2GT models global, patch, and pixel representations and their hierarchical relations (Feng et al., 2025a).

Structure-aware methods further exploit local dependencies and relational patterns. HOS-Net combines short- and long-range features with hypergraph-based high-order structure learning and middle-feature alignment (Qiu et al., 2024). Polymorphic masking and wavelet graph convolution use global and structural branches to improve robustness under corrupted inputs (Sun et al., 2024). HHGF models within-modality homogeneous relations and cross-modal heterogeneous correlations between local features (Feng et al., 2025b). For occluded VI-ReID, an occlusion-aware completion network reconstructs missing cues at token and region levels before alignment (Yu et al., 2026). These approaches broaden the spatial, structural, and semantic evidence available for matching, while the resulting integration remains closely related to how its input features are organized.

Taken together, feature disentanglement organizes identity- and modality-related information, cross-modal alignment regulates consistency between visible and infrared representations, and attention-based multi-granularity modeling selects and combines cues across regions and levels. In many existing methods, functional feature organization and subsequent alignment or interaction are nevertheless treated as relatively separate design stages. This motivates the decoupling-guided alignment strategy studied in this work: MBUE organizes complementary global-relation and local-spatial feature streams with different empirical information tendencies, and DIG-CMA performs guided cross-stream integration. Under the joint optimization objective, the resulting representations support cross-modal representation alignment.

3. Proposed method

As shown in Fig. 2, DIGCA comprises a Hierarchical Context Extractor (HCE), a Multi-Branch Unified Encoder (MBUE), and a Decoupled Information-Guided Cross-Modal Alignment (DIG-CMA) module. HCE enriches the backbone features with multi-scale contextual information. Within MBUE, parallel global-relation and local-spatial modeling organizes complementary feature streams, which are refined and integrated by DIG-CMA through guided cross-stream interaction. The resulting representations are jointly optimized using the objective described in Section 3.4 to support cross-modal representation alignment.

3.1. HCE: multi-scale context learning

To improve multi-scale feature perception and capture information ranging from local details to global context, we design a Hierarchical Context Extractor (HCE) inspired by STDC (Fan et al., 2021) and DSNet (Guo et al., 2025b). HCE jointly encodes pedestrian foreground information and its surrounding context. Together with the local representation capability of ResNet, HCE constructs a richer identity representation.

As illustrated in Fig. 2, HCE consists of three dilated convolutional layers with different dilation rates to enlarge the receptive field (RF). The receptive field of a single dilated convolutional layer is calculated as follows:

$$RF = k + (k - 1) \times (d - 1), \tag{1}$$

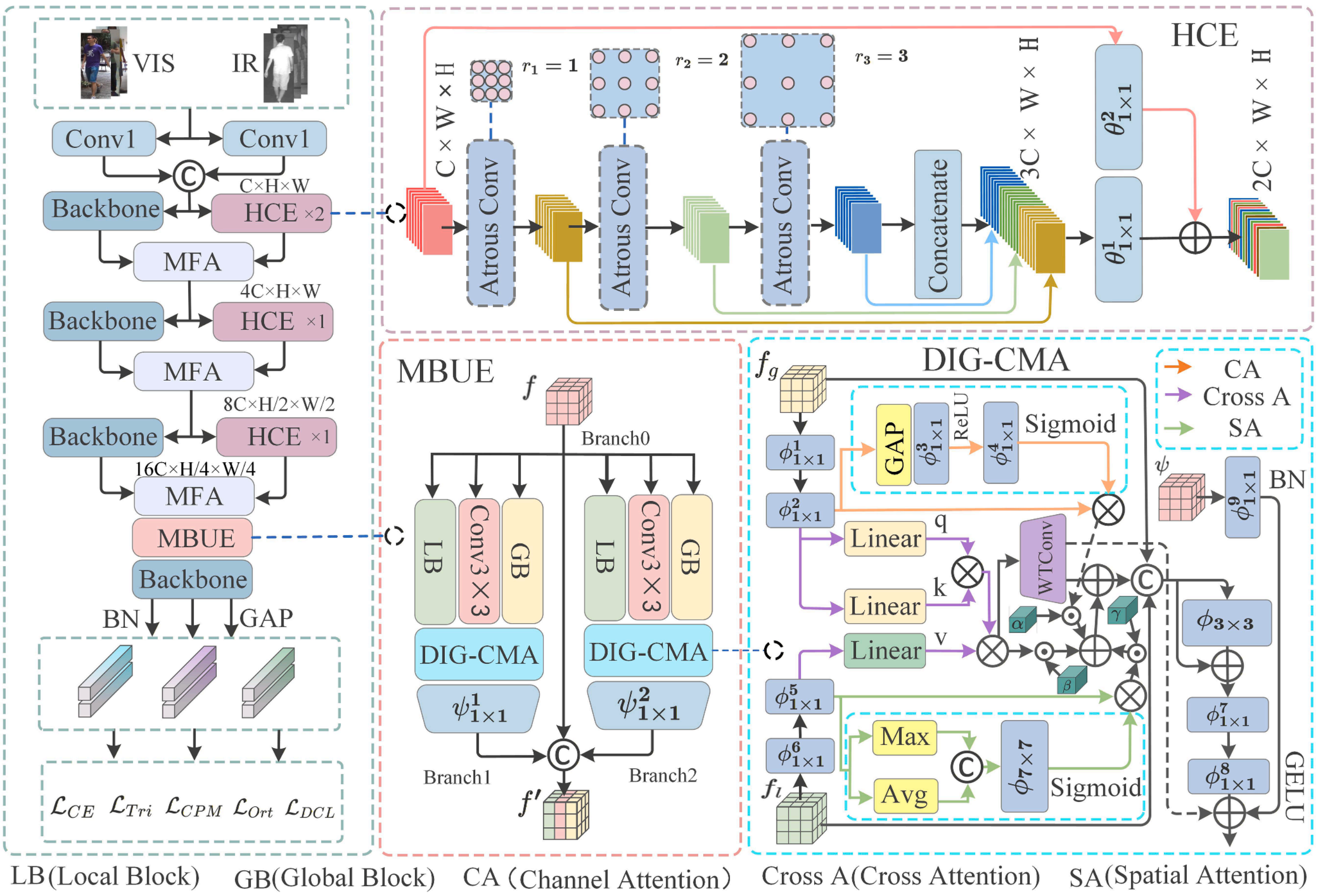


Fig. 2. Overall architecture of DIGCA. HCE and the backbone network (He et al., 2016) aggregate multi-scale contextual information to enrich the initial identity representation; MFA (Zhang & Wang, 2023) performs feature interaction; MBUE models global relations and local spatial information in parallel to organize complementary representations; and DIG-CMA uses functionally differentiated representations to guide cross-stream interaction for cross-modal representation alignment.

where k denotes the kernel size and d denotes the dilation rate.

When multiple dilated convolution layers are stacked, the calculation method for the receptive field is as follows:

$$RF_n = RF_{n-1} + (k_n - 1) \times (d_n - 1) \times RF_{n-1}, \tag{2}$$

where n is the number of stacked dilated convolutional layers, RF_{n-1} denotes the receptive field of the $(n-1)$th layer, and k_n and d_n denote the kernel size and dilation rate of the nth layer, respectively.

As indicated by Eqs. (1) and (2), different dilation rates allow the layers to cover contextual information at different scales. On this basis, HCE incorporates residual connections to aggregate multi-scale contextual information and enrich the initial identity features extracted by ResNet.

3.2. Multi-branch unified encoder

Following the decoupling-guided alignment strategy, we construct a Multi-Branch Unified Encoder (MBUE) with two separately parameterized parallel fusion paths. Each path derives local-spatial, global-relation, and additional convolutional features from the same input, and a dedicated DIG-CMA unit is nested within the path to refine and integrate these complementary features. The resulting streams are not assigned fixed modality-stable or modality-sensitive roles; their functional tendencies are evaluated empirically in Section 4.5.

Specifically, MBUE contains two separately parameterized parallel fusion paths. Each path comprises a local block, a global block, a plain 3×3 convolution, and a dedicated DIG-CMA unit. The resulting local-spatial, global-relation, and convolutional features are supplied to the corresponding DIG-CMA unit for guided integration, after which the fused output is projected back to the input channel dimension. The original input is retained as Branch0, while the fusion outputs of the first and second paths form Branch1 and Branch2, respectively. Together, they constitute three complementary feature streams.

Inspired by Swin Transformer (Liu et al., 2021), the global block employs shifted-window multi-head self-attention (SW-MSA) to model feature relations within shifted window partitions. Layer normalization is applied before both the attention and MLP operations, and each operation is followed by a residual connection with DropPath. Let $\mathrm{DP}(\cdot)$ denote DropPath. The global-relation feature is computed as

$$\begin{aligned} \hat{f}_g &= f + \mathrm{DP}(\text{SW-MSA}(\mathrm{LN}(f))), \\ f_g &= \hat{f}_g + \mathrm{DP}(\mathrm{MLP}(\mathrm{LN}(\hat{f}_g))) \end{aligned} \tag{3}$$

where f denotes the input feature and $LN(\cdot)$ denotes LayerNorm. The shifted-window design facilitates relation modeling across neighboring window partitions, without assuming that a single attention operation covers the entire feature map.

The local block models spatial patterns using a 3×3 depthwise convolution followed by channels-last LayerNorm, a channel-wise linear projection, GELU activation, and a residual connection. The linear projection is applied independently at each spatial location and is equivalent to a pointwise channel mapping. Its operation is written as

$$f_l = f + \mathrm{DP}\big(\mathrm{GELU}\big(W_p\big(\mathrm{LN}\big(D_{3\times3}(f)\big)\big)\big)\big), \tag{4}$$

where $D_{3\times3}(\cdot)$ denotes depthwise convolution and $W_p(\cdot)$ denotes the channel-wise linear projection. In the implementation, the feature is temporarily arranged in channels-last format for LayerNorm and W_p, and is restored to channels-first format before the residual addition.

The independently parameterized paths provide complementary inputs for the guided cross-stream integration performed by their nested DIG-CMA units. Their outputs, together with the original input stream, form representations with different empirical functional tendencies. MBUE therefore promotes partial functional decoupling rather than a strict separation of modality-shared and modality-specific information. The three streams are used by the joint objective in Section 3.4 to support cross-modal identity matching and representation alignment.

3.3. Decoupled information-guided cross-modal alignment

To coordinate the complementary information captured by global-relation and local-spatial modeling, MBUE incorporates two separately parameterized DIG-CMA units to construct two parallel fusion streams. Each DIG-CMA unit first aligns the channel dimensions of the two input streams, refines them using channel and spatial attention, respectively, and then integrates their information through cross-attention and feature enhancement. The detailed operations are described below.

First, given a global feature map $f_g \in \mathbb{R}^{B\times C_1\times H\times W}$ and a local feature map $f_l \in \mathbb{R}^{B\times C_2\times H\times W}$, two 1×1 convolutions are applied to each feature map to align them to the same dimensionality, as follows:

$$f_g^a = \phi_{1\times1}^2(\phi_{1\times1}^1(f_g)), \tag{5}$$

$$f_l^a = \phi_{1\times1}^4(\phi_{1\times1}^3(f_l)), \tag{6}$$

where f_g^a and f_l^a denote the dimensionally aligned global and local feature maps, respectively.

Next, channel attention is applied to the global feature to capture inter-channel semantic dependencies and emphasize informative global semantics. Spatial attention is applied to the local feature to highlight salient spatial regions and suppress irrelevant background information. The operations are formulated as follows:

$$M_c = \sigma(\phi_{1\times1}^5(ReLU(\phi_{1\times1}^6(Avg(f_g^a))))), \tag{7}$$

$$F_c = f_g^a \otimes M_c, \tag{8}$$

$$M_s = \sigma(\phi_{7\times7}(Cat_{d=1}(Max(f_l^a), Avg(f_l^a)))), \tag{9}$$

$$F_s = f_l^a \otimes M_s, \tag{10}$$

In Eq. (7), the spatial dimensions of f_g^a are first compressed by global average pooling $Avg(\cdot)$. The 1×1 convolution $\phi_{1\times1}^6$ then reduces the channel dimensionality, and $ReLU(\cdot)$ introduces nonlinearity. Subsequently, the 1×1 convolution $\phi_{1\times1}^5$ restores the original number of channels, and $\sigma(\cdot)$ generates the channel-attention weights M_c. In Eq. (8), f_g^a is multiplied channel-wise by M_c through $\otimes$ to obtain the attended global feature F_c.

Similarly, in Eq. (9), global max pooling $Max(\cdot)$ and global average pooling $Avg(\cdot)$ are applied to f_l^a, and the resulting features are concatenated along dimension $d = 1$. A 7×7 convolution $\phi_{7\times7}$ then performs spatial transformation, followed by the Sigmoid function $\sigma(\cdot)$ to generate the spatial-attention weights M_s. Finally, as shown in Eq. (10), f_l^a is multiplied by M_s at each spatial location to obtain the attended local feature F_s.

Cross-attention is subsequently used to capture semantic correlations between the two feature types and facilitate their interaction, as follows:

$$F_{cross} = softmax(\frac{qk^T}{\sqrt{d_k}})v, \tag{11}$$

where $q = W_q f_g^a$, $k = W_k f_l^a$, and $v = W_v f_l^a$. Here, W_q, W_k, and W_v are learnable projection matrices that map the original features into different representation spaces.

The outputs of the three attention mechanisms are then dynamically integrated using the learnable weights α, β, and γ, allowing their contributions to be adjusted according to the task and data, as follows:

$$\begin{aligned} &F_{fused} = \alpha F_{cross} + \beta F_s + \gamma F_c, \\ &\text{where } \alpha+\beta+\gamma = 1,\ \alpha,\beta,\gamma > 0, \end{aligned} \tag{12}$$

where α, β, and γ are obtained by applying a softmax function to three unconstrained learnable parameters.

Finally, wavelet transform convolution (WTConv2d) is applied after attention-based cross-stream integration to enhance the feature representation, as follows:

$$F_{enhanced} = F_{fused} + WTConv(F_{fused}), \tag{13}$$

Through this process, each DIG-CMA unit produces an enhanced fusion stream while preserving information from the original global-relation and local-spatial streams through the final fusion and residual connection. The two parallel fusion streams, together with the original MBUE stream, are jointly optimized using the objective described in Section 3.4 to support cross-modal representation alignment.

3.4. Joint optimization objective

We retain the identity-classification, hard-mined triplet (Hermans et al., 2017), center-guided pair mining (CPM), and orthogonality losses adopted from DEEN (Zhang & Wang, 2023). In addition, discriminative center loss (DCL) (Lu et al., 2023) is applied to the three output streams to enhance their class-level compactness. The individual loss terms are described below.

For identity supervision, we apply cross-entropy loss to all three feature streams:

$$\mathcal{L}_{CE} = -\frac{1}{M}\sum_{b=1}^{M}\sum_{i=1}^{N} y_{b,i}\log p_{b,i}, \tag{14}$$

where M is the number of samples, N is the number of identities, and $p_{b,i}$ and $y_{b,i}$ are the predicted probability and one-hot target, respectively.

We further use batch-hard triplet loss to improve identity separation:

$$\mathcal{L}_{Tri} = \frac{1}{M}\sum_{a=1}^{M}\left[d(f_a, f_{p(a)}) - d(f_a, f_{n(a)}) + m_t\right]_+, \tag{15}$$

where $p(a)$ and $n(a)$ are the hardest positive and negative samples for anchor a, respectively; $d(\cdot,\cdot)$ is the Euclidean distance, $m_t = 0.3$, and $[z]_+ = \max(z, 0)$.

CPM imposes bidirectional ranking constraints on modality-specific and generated-stream class centers:

$$\begin{aligned} \mathcal{L}_{CPM} = &\left[D(\boldsymbol{c}_n^i, \boldsymbol{c}_{v+}^i) - D(\boldsymbol{c}_v^j, \boldsymbol{c}_{v+}^i) - D(\boldsymbol{c}_v^j, \boldsymbol{c}_v^k) + m_c\right]_+ \\ &+ \left[D(\boldsymbol{c}_v^j, \boldsymbol{c}_{n+}^i) - D(\boldsymbol{c}_n^j, \boldsymbol{c}_{n+}^i) - D(\boldsymbol{c}_n^j, \boldsymbol{c}_n^k) + m_c\right]_+, \end{aligned} \tag{16}$$

$$\begin{aligned} \mathcal{L}_{CPM} = &[D(c_n^i, c_{v+}^i) - D(c_v^j, c_{v+}^i) - D(c_v^j, c_v^k) + m_c]_+ \\ &+ [D(c_v^j, c_{n+}^i) - D(c_n^j, c_{n+}^i) - D(c_n^j, c_n^k) + m_c]_+ \end{aligned} \tag{17}$$

where $\boldsymbol{c}_v$ and $\boldsymbol{c}_n$ are visible and infrared class centers, $\boldsymbol{c}_{v+}$ and $\boldsymbol{c}_{n+}$ are the corresponding generated-stream centers, and j and k denote the mined positive and negative center indices, respectively. The implementation uses $m_c = 0.2$. CPM is evaluated between the original stream and each of the two MBUE streams, and the two values are averaged.

The inherited orthogonality term is applied to the pooled outputs of the two MBUE-generated streams:

$$\mathcal{L}_{Ort} = \frac{1}{3B}\sum_{b=1}^{B}\left(\mathbf{z}_b^{(1)}\right)^T\mathbf{z}_b^{(2)}, \tag{18}$$

where B is the number of visible–infrared samples before the three-stream expansion, and $\mathbf{z}_b^{(1)}$ and $\mathbf{z}_b^{(2)}$ are the pooled representations of the two generated streams. This term follows the DEEN implementation and discourages redundant responses between the generated representations.

For DCL, a mini-batch contains P identities and K samples per modality for each identity. The center of identity u_i is computed from its $2K$ visible and infrared samples. Let $\mathcal{P}$ denote all nonzero center-to-sample distances for samples of the same identity, and let $\mathcal{N}_i$ denote distances from center c_{u_i} to samples of different identities. Hard negatives are selected as $\mathcal{H}_i = \{k \in \mathcal{N}_i : \|f_k - c_{u_i}\|_2 < \bar{d_i^-}\}$,

Table 1
Comparison with state-of-the-art methods on SYSU-MM01. For DIGCA, each metric is averaged over three independent training runs (seeds 0, 1, and 2). Cross-run standard deviations for R1 and mAP are reported in Table 4.

Method	All Search				Indoor Search			
	mAP	R1 (%)	R10 (%)	R20 (%)	mAP	R1 (%)	R10 (%)	R20 (%)
AGW (Ye et al., 2021)	47.6	47.5	N/A	N/A	62.9	54.1	N/A	N/A
NFS (Chen et al., 2021)	55.4	56.9	91.3	96.5	69.7	62.7	96.5	99.0
MPANet (Wu et al., 2021)	68.2	70.5	96.2	98.8	80.9	76.7	98.2	99.5
MCLNet (Hao et al., 2021)	61.9	65.4	93.3	97.1	76.5	72.5	96.9	99.2
SMCL (Wei et al., 2021)	61.7	67.3	92.8	96.7	75.5	68.8	96.5	95.2
FMCNet (Zhang et al., 2022)	62.5	66.3	N/A	N/A	74.0	68.1	N/A	N/A
DART (Yang et al., 2022)	66.2	68.7	96.3	98.9	78.1	72.5	97.8	99.4
MAUM^P (Liu et al., 2022b)	68.7	71.6	N/A	N/A	81.9	76.9	N/A	N/A
PMT (Lu et al., 2023)	64.9	67.5	95.3	95.3	76.5	71.6	96.7	99.2
MRCN (Zhang et al., 2023)	65.5	68.9	95.2	98.4	79.8	76.0	98.3	99.7
DEEN (Zhang & Wang, 2023)	71.8	74.7	97.6	99.2	83.3	80.3	99.0	99.8
SEFEL (Feng et al., 2023a)	70.1	75.2	96.9	99.1	81.2	78.4	97.5	98.9
PartMix (Kim et al., 2023)	74.6	77.8	N/A	N/A	84.4	81.5	N/A	N/A
IDKL^a (Ren & Zhang, 2024)	68.5	70.8	95.6	98.3	82.7	79.9	97.7	99.3
CAJ (Ye et al., 2024)	68.1	71.4	96.2	98.7	81.9	78.3	98.3	99.7
TMD (Lu et al., 2024)	67.7	73.9	96.2	98.7	78.8	81.1	98.8	99.6
DCPLNet (Chan et al., 2024)	70.4	74.0	96.5	98.9	81.9	78.3	98.7	99.8
HOS-Net (Qiu et al., 2024)	74.2	75.6	N/A	N/A	86.7	84.2	N/A	N/A
MHN (Zuo et al., 2025)	71.8	75.0	97.2	N/A	85.0	81.5	98.8	N/A
SCR (Yu et al., 2025)	71.2	75.5	97.7	99.4	86.1	84.7	99.1	99.9
AMML (Zhang et al., 2025)	74.8	77.8	97.8	99.4	88.3	86.6	**99.5**	**100.0**
MSCMNet (Hua et al., 2025)	74.2	78.5	97.5	99.2	85.5	83.0	98.9	99.8
DEEN+DiVE (Dai et al., 2025)	74.9	79.0	N/A	N/A	85.9	82.9	N/A	N/A
DIGCA (Ours)	**76.4**	**79.3**	**98.3**	**99.5**	**88.8**	**87.2**	99.1	99.7

where $\bar{d}_i^-$ is the mean of all negative distances for identity u_i. DCL is

$$\mathcal{L}_{DCL} = \frac{\frac{1}{|\mathcal{P}|}\sum_{(i,j)\in\mathcal{P}} \|f_j - c_{u_i}\|_2}{\frac{1}{P}\sum_{i=1}^{P}\frac{1}{|\mathcal{H}_i|}\sum_{k\in\mathcal{H}_i}\|f_k - c_{u_i}\|_2}. \tag{19}$$

The numerator measures intra-class compactness, whereas the denominator measures the separation from hard inter-class samples. DCL is computed independently for the original stream and the two MBUE-generated streams without feature normalization.

The complete objective is

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \mathcal{L}_{Tri} + \lambda_1\mathcal{L}_{CPM} + \lambda_2\mathcal{L}_{Ort} + \lambda_3\sum_{r=0}^{2}\mathcal{L}_{DCL}^{(r)} \tag{20}$$

where $r = 0$ denotes the original stream and $r = 1, 2$ denote the two MBUE-generated streams. Cross-entropy and triplet losses are evaluated over all three streams jointly; CPM links the original stream with each generated stream; and the orthogonality term acts between the two generated streams. The loss weights used in the experiments are reported in the implementation details.

Overall, the joint objective constrains the learned representations in terms of identity discrimination, metric structure, feature complementarity, and class-level compactness, thereby supporting cross-modal identity matching and representation alignment.

4. Experiments

4.1. Experimental settings

4.1.1. Datasets and evaluation protocols

We evaluate DIGCA on three visible–infrared person re-identification benchmarks: SYSU-MM01 (Wu et al., 2017), RegDB (Du et al., 2018), and LLCM (Zhang & Wang, 2023). Following the standard protocol of each dataset, cumulative matching characteristic (CMC) accuracy and mean average precision (mAP) are adopted as evaluation metrics. R1, R10, and R20 denote the retrieval accuracies at ranks 1, 10, and 20, respectively.

SYSU-MM01 contains 303,420 images of 491 identities captured by four visible and two infrared cameras. The official split uses 395 identities for training and 96 identities for testing. We report results under the All Search and Indoor Search settings. For each setting, the gallery is sampled 10 times following the standard single-shot protocol, and performance is averaged over these gallery trials. This protocol evaluates retrieval under different indoor and outdoor conditions.

RegDB contains 8240 images of 412 identities, with 10 visible and 10 infrared images per identity. Following the standard protocol, 206 identities are used for training and the remaining 206 for testing. We report results for visible-to-infrared (VIS-to-IR) and infrared-to-visible (IR-to-VIS) retrieval, averaged over 10 random training/test splits.

LLCM is a low-light cross-modal benchmark collected by nine cameras. It contains 46,767 images of 1064 identities, including 713 identities for training and 351 for testing. Following the official evaluation protocol, we report results for both VIS-to-IR and IR-to-VIS retrieval under challenging low-light imaging conditions.

4.1.2. Implementation details

DIGCA is implemented in PyTorch and trained on a single NVIDIA RTX A6000 GPU. A ResNet-50 (He et al., 2016) pretrained on ImageNet (Deng et al., 2009) is used as the backbone. Input images are resized to 384×192 pixels. During training, data augmentation is applied sequentially using 10-pixel padding, random cropping, random grayscale conversion with a probability of 0.5, random horizontal flipping, and random erasing with a probability of 0.5, followed by ImageNet normalization. The model is optimized for 150 epochs using stochastic gradient descent with Nesterov momentum, where the momentum is 0.9 and the weight decay is 5×10^{-4}. The initial learning rate is set to 0.1 for the bottleneck and classifier layers and 0.01 for the remaining network pa-

Table 2
Comparison with state-of-the-art methods on RegDB.

Methods	VIS to IR		IR to VIS	
	R1 (%)	mAP	R1 (%)	mAP
AGW (Ye et al., 2021)	70.0	66.3	70.4	65.9
NFS (Chen et al., 2021)	80.5	72.1	78.0	69.8
MPANet (Wu et al., 2021)	82.8	80.7	83.7	80.9
MCLNet (Hao et al., 2021)	80.3	73.1	75.9	69.5
SMCL (Wei et al., 2021)	83.9	79.8	83.0	78.5
DART (Yang et al., 2022)	83.6	75.7	82.0	73.8
MAUM^{P} (Liu et al., 2022b)	87.8	85.0	86.9	<u>84.3</u>
FMCNet (Zhang et al., 2022)	89.1	84.4	88.4	83.2
PMT (Lu et al., 2023)	67.5	64.9	71.6	76.5
MRCN (Zhang et al., 2023)	91.4	84.6	88.3	81.9
CMTR (Liang et al., 2023)	88.1	81.6	84.9	80.8
DEEN (Zhang & Wang, 2023)	91.1	<u>85.1</u>	<u>89.5</u>	83.4
PartMix (Kim et al., 2023)	85.7	82.3	84.9	82.5
CM^{2}GT (Feng et al., 2025a)	86.7	77.7	86.4	77.5
MSFCS (Yang et al., 2024)	85.3	76.3	83.8	75.1
CFET (Guo et al., 2025a)	87.0	83.6	86.8	84.2
DMPF (Lu et al., 2025)	88.8	81.0	88.8	81.8
AGPI^{2} (Alehdaghi et al., 2025)	89.0	83.8	87.9	83.0
PMWGCN (Sun et al., 2024)	90.6	84.5	88.7	81.6
DARD (Wei et al., 2024)	91.1	<u>85.1</u>	<u>89.5</u>	83.4
TMD (Lu et al., 2024)	**93.0**	84.3	87.4	81.3
DIGCA (Ours)	<u>92.0</u>	**85.3**	**91.1**	**84.5**

Table 3
Comparison with state-of-the-art methods on LLCM.

Methods	VIS to IR		IR to VIS	
	R1 (%)	mAP	R1 (%)	mAP
AGW (Ye et al., 2021)	49.1	55.8	63.7	47.2
DART (Yang et al., 2022)	52.9	59.2	65.3	51.1
MRCN (Zhang et al., 2023)	51.3	57.7	65.3	49.5
CAJ^{1} (Ye et al., 2021a)	49.8	56.4	63.7	47.7
CM^{2}GT (Feng et al., 2025a)	52.1	58.3	65.9	50.3
MFLNet (Yang et al., 2025)	53.5	59.7	67.2	51.9
DEEN (Zhang & Wang, 2023)	55.5	62.1	69.2	<u>55.5</u>
MHN (Zuo et al., 2025)	<u>56.6</u>	<u>64.4</u>	<u>69.7</u>	55.1
AMML (Zhang et al., 2025)	54.3	60.8	68.3	53.4
DIGCA (Ours)	**57.9**	**64.6**	**70.5**	**57.5**

rameters. Each mini-batch contains six identities, with four visible and four infrared images sampled for each identity. Unless otherwise stated, λ_1 and λ_2 in Eq. (20) are set to 0.8 and 0.01, respectively. The DCL weight λ_3 is fixed at 0.1 throughout training. All compared variants use the same data split, input resolution, optimizer, identity-sampling strategy, and training schedule.

4.2. Comparison with state-of-the-art methods

Tables 2–3 compare DIGCA with recent VI-ReID methods on SYSU-MM01, RegDB, and LLCM. Bold and underlined values indicate the best and second-best results, respectively. Overall, DIGCA achieves competitive performance on all three benchmarks, with more apparent advantages on SYSU-MM01 and LLCM.

SYSU-MM01. As shown in Table 1, DIGCA obtains 79.3% R1 and 76.4% mAP under All Search, and 87.2% R1 and 88.8% mAP under Indoor Search. Under All Search, DIGCA exceeds TMD (Lu et al., 2024) by 5.4 percentage points in R1 and 8.7 percentage points in mAP. Compared with MHN (Zuo et al., 2025), the corresponding improvements are 4.3 and 4.6 percentage points. DIGCA also outperforms AMML (Zhang et al., 2025) by 1.5 percentage points in R1 and 1.6 percentage points in mAP, while the gains over DEEN+DiVE (Dai et al., 2025) are 0.3 and 1.5 percentage points, respectively. Compared with HOS-Net (Qiu et al., 2024), DIGCA improves R1 by 3.7 percentage points and mAP by 2.2 percentage points. DIGCA also improves upon HOS-Net (Qiu et al., 2024), which employs high-order structural modeling, by 3.7 percentage points in R1 and 2.2 percentage points in mAP. Under Indoor Search, AMML remains the closest competing method, obtaining 86.6% R1 and 88.3% mAP. These results indicate that organizing complementary feature streams before guided integration benefits retrieval under the multi-camera conditions of SYSU-MM01. The improvements are mainly reflected in R1 and mAP, while the R10 and R20 results of several recent methods are already close to saturation.

RegDB. Table 2 shows that DIGCA obtains 92.0% R1 and 85.3% mAP for VIS-to-IR retrieval, and 91.1% R1 and 84.5% mAP for IR-to-VIS retrieval. Its VIS-to-IR mAP and IR-to-VIS R1 and mAP are among the stronger results in the comparison, although its VIS-to-IR R1 remains 1.0 percentage point below that of TMD (Lu et al., 2024). Compared with DEEN (Zhang & Wang, 2023), the improvement in the IR-to-VIS direction is modest, suggesting that the additional benefit of decoupling-guided alignment may be limited under the relatively controlled acquisition conditions of RegDB. By contrast, the improvements are more evident on SYSU-MM01 and LLCM, which involve more diverse camera and illumination conditions. Overall, DIGCA maintains competitive performance in both retrieval directions on RegDB.

LLCM. As shown in Table 3, DIGCA obtains 57.9% R1 and 64.6% mAP for VIS-to-IR retrieval, and 70.5% R1 and 57.5% mAP for IR-to-VIS retrieval. In the VIS-to-IR direction, DIGCA improves upon MHN (Zuo et al., 2025), the strongest competing method in this setting, by 1.3 percentage points in R1 and 0.2 percentage points in mAP. In the reverse direction, it exceeds the highest R1 among the compared methods, also obtained by MHN, by 0.8 percentage points and the highest mAP, obtained by DEEN (Zhang & Wang, 2023), by 2.0 percentage points. The improvements are observed in both retrieval directions, although their magnitude remains modest for some metrics.

4.3. Stability analysis

To assess the sensitivity of DIGCA to training randomness, we independently train the complete model using random seeds 0, 1, and 2. Table 4 reports the stability results on SYSU-MM01.

As shown in Table 4, DIGCA obtains 79.33±0.41% R1 and 76.37±0.24% mAP under All Search. Under Indoor Search, the corresponding results are 87.19±0.46% and 88.81±0.34%, respectively. The standard deviations are below 0.60 percentage points in both settings, indicating that the reported performance is reproducible across different random initializations.

4.4. Ablation study

To systematically evaluate the proposed design, we conduct ablation experiments on SYSU-MM01 under both All Search and Indoor Search. We first examine the respective contributions of HCE, MBUE, DIG-CMA, and DCL, and then investigate the effects of WTConv, cross-attention, and global–local branch configurations. Finally, we analyze sensitivity to the DCL weight λ_3. Unless otherwise stated, all variants use the same training and evaluation settings.

Effect of HCE. As shown in Table 5, incorporating HCE into the DEEN baseline increases R1 from 74.7% to 75.6% and mAP from 71.8% to 72.6% under All Search. Under Indoor Search, R1 increases from 80.3% to 81.6%, while mAP increases from 83.3% to 84.5%. The consistent improvements under both settings are in line with the intended role of HCE in enriching the backbone representation through multi-scale contextual aggregation.

Effect of MBUE. To isolate the contribution of MBUE, we incorporate it into the DEEN baseline without using DIG-CMA. Under All Search, R1 increases from 74.7% to 75.0%, while mAP increases from 71.8% to 72.3%. Under Indoor Search, R1 and mAP increase from 80.3% and 83.3% to 82.4% and 85.0%, corresponding to gains of 2.1 and 1.7 percentage points, respectively. These results show that the parallel global-

Table 4
Stability of DIGCA on SYSU-MM01 over three independent training runs (seeds 0, 1, and 2). Each run is averaged over 10 standard gallery trials; mean±std is computed across the three runs.

Setting	Seed 0		Seed 1		Seed 2		Mean±std	
	R1	mAP	R1	mAP	R1	mAP	R1	mAP
All Search	79.51	76.48	78.87	76.09	79.62	76.54	79.33±0.41	76.37±0.24
Indoor Search	87.64	89.09	86.72	88.44	87.21	88.91	87.19±0.46	88.81±0.34

Table 5
Component ablation results on SYSU-MM01 using a fixed random seed (seed 0).

Baseline	HCE	MBUE	DIG-CMA	DCL	All Search		Indoor Search	
					R1 (%)	mAP	R1 (%)	mAP
✓					74.7	71.8	80.3	83.3
✓	✓				75.6	72.6	81.6	84.5
✓		✓			75.0	72.3	82.4	85.0
✓		✓	✓		76.7	74.0	84.8	86.9
✓		✓	✓	✓	77.8	74.4	85.6	87.5
✓	✓	✓	✓	✓	**79.5**	**76.4**	**87.6**	**89.0**

Table 6
Ablation of WTConv on SYSU-MM01 using a fixed random seed (seed 0).

WTConv	All Search				Indoor Search			
	R1 (%)	R10 (%)	R20 (%)	mAP	R1 (%)	R10 (%)	R20 (%)	mAP
No	78.71	97.91	99.30	75.79	86.81	99.45	99.93	88.44
Yes	79.51	98.53	99.72	76.48	87.64	99.30	99.98	89.09
Δ	**0.80**	**0.62**	**0.42**	**0.69**	**0.83**	−0.15	**0.05**	**0.65**

relation and local-spatial modeling introduced by MBUE improves retrieval performance even without the guided cross-stream integration performed by DIG-CMA.

Effect of DIG-CMA. Incorporating DIG-CMA into MBUE increases R1 and mAP from 75.0% and 72.3% to 76.7% and 74.0% under All Search. Under Indoor Search, R1 increases from 82.4% to 84.8%, while mAP increases from 85.0% to 86.9%. These results show that refining the global-relation and local-spatial streams through channel and spatial attention and integrating them through cross-attention further improves retrieval performance over MBUE alone. DIG-CMA therefore enables the complementary feature streams organized by MBUE to be jointly used for subsequent cross-modal representation alignment.

Effect of DCL. The identity-classification, triplet, CPM, and orthogonality losses are adopted unchanged from DEEN and have been evaluated in the original study. We therefore keep these losses fixed and isolate the contribution of the newly introduced DCL. With MBUE and DIG-CMA retained, adding DCL increases R1 from 76.7% to 77.8% and mAP from 74.0% to 74.4% under All Search. Under Indoor Search, R1 increases from 84.8% to 85.6%, while mAP increases from 86.9% to 87.5%. DCL thus provides an additional class-level constraint that improves retrieval performance in both search settings. After HCE is further included, the complete configuration achieves the highest overall results in Table 5, showing that the proposed components are effective when used together.

Effect of WTConv. As shown in Table 6, WTConv improves R1 and mAP by 0.80 and 0.69 percentage points under All Search, and by 0.83 and 0.65 percentage points under Indoor Search, respectively. These results show that WTConv provides a modest improvement in the principal retrieval metrics by enhancing the features after attention-based integration. However, the 0.15-percentage-point decrease in Indoor Search R10 indicates that its benefit does not extend uniformly to every retrieval rank.

Effect of cross-attention. The variant without cross-attention follows the DEEN baseline and aggregates the parallel projections using an unweighted arithmetic mean. As shown in Table 7, replacing mean aggregation with cross-attention improves R1 and mAP by 1.62 and 1.23 percentage points under All Search, and by 1.15 and 0.77 percentage points under Indoor Search, respectively. These gains show that cross-attention allows DIG-CMA to integrate the global-relation and local-spatial streams according to their learned feature correlations, providing more targeted cross-stream integration than fixed mean aggregation.

Effect of branch configurations. Table 8 compares the global-only, local-only, and combined global–local configurations. Under All Search, combining the two feature sources improves R1 and mAP by 1.10 and 2.23 percentage points over the global-only configuration, and by 1.89 and 3.45 percentage points over the local-only configuration, respectively. The combined configuration also achieves the highest R1 and mAP under Indoor Search. The results show that global-relation and local-spatial modeling provide complementary cues, and their combination produces more effective features for cross-modal identity matching.

Hyperparameter analysis. Within the evaluated range and the current training protocol, we vary the DCL weight λ_3 while keeping all other settings fixed. As shown in Fig. 3, $\lambda_3 = 0.1$ achieves the highest observed performance, whereas both smaller and larger values produce lower results. This variation shows that the joint objective is sensitive to the relative contribution of DCL. Accordingly, we set λ_3 to 0.1 in the remaining experiments.

4.5. *Analysis of functionally differentiated MBUE features*

To assess the functional differentiation of MBUE, we analyze three BN-normalized feature streams extracted from the seed-0 model on SYSU-MM01. Branch0 denotes the original MBUE input stream, while Branch1 and Branch2 are produced by two separately parameterized parallel fusion streams, each combining global-relation and local-spatial features. We compare their empirical behavior without assigning fixed global-only or local-only roles to individual streams. The analysis includes branch-wise retrieval, an identity-disjoint linear modality probe, cross-modal similarity and linear-dependence statistics, and t-SNE visualization.

As shown in Table 9, Branch0, Branch1, and Branch2 achieve comparable identity-retrieval performance, with Branch1 and Branch2 slightly

Table 7
Ablation of cross-attention in DIG-CMA on SYSU-MM01 using a fixed random seed (seed 0).

Cross-attention	All Search				Indoor Search			
	R1 (%)	R10 (%)	R20 (%)	mAP	R1 (%)	R10 (%)	R20 (%)	mAP
No	77.89	97.61	99.52	75.25	86.49	99.07	99.84	88.32
Yes	79.51	98.53	99.72	76.48	87.64	99.30	99.98	89.09
Δ	**1.62**	**0.92**	**0.20**	**1.23**	**1.15**	**0.23**	**0.14**	**0.77**

Table 8
Comparison of global and local MBUE feature sources on SYSU-MM01 using a fixed random seed (seed 0).

Feature source	All Search				Indoor Search			
	R1 (%)	R10 (%)	R20 (%)	mAP	R1 (%)	R10 (%)	R20 (%)	mAP
Global-only	78.41	97.66	99.53	74.25	86.51	99.16	99.86	88.10
Local-only	77.62	97.50	99.42	73.03	86.13	99.46	99.97	87.84
Global + Local	**79.51**	**98.53**	**99.72**	**76.48**	**87.64**	**99.30**	**99.98**	**89.09**

Table 9
Branch-wise retrieval on SYSU-MM01 using the seed-0 model. BN Sum combines the three BN-normalized streams, whereas full DEEN-style inference additionally includes the three pooled streams. Values are mean±SD over 10 gallery trials, with SD reflecting gallery-sampling variation.

Feature	All Search			Indoor Search		
	R1 (%)	R20 (%)	mAP	R1 (%)	R20 (%)	mAP
Branch0	77.82±2.11	99.38±0.17	75.05±1.51	85.23±1.92	**99.80±0.31**	87.27±1.37
Branch1	78.14±1.58	99.43±0.20	75.14±1.42	85.68±1.73	99.58±0.52	87.38±1.41
Branch2	78.06±1.78	99.40±0.18	75.21±1.44	85.62±1.60	99.64±0.43	87.37±1.28
Branch0+Branch2	78.51±1.93	99.40±0.16	75.73±1.45	86.04±1.66	99.76±0.33	87.78±1.27
BN_Sum	**78.72±1.73**	**99.44±0.17**	**75.83±1.45**	**87.11±1.71**	99.70±0.37	**88.81±1.37**

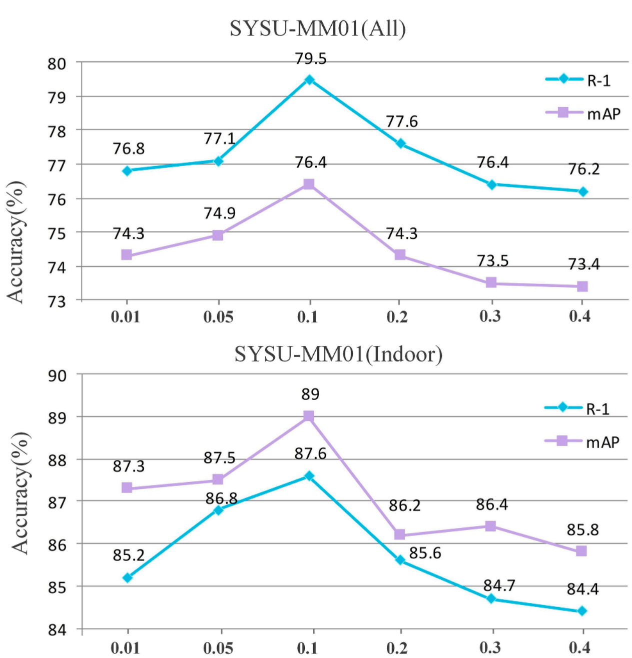


Fig. 3. Influence of different λ_3 on the results of our method on the SYSU-MM01 dataset.

exceeding Branch0 in both R1 and mAP. Combining Branch0 with Branch2 further improves the results, while BN Sum achieves the highest R1 and mAP under both search settings. The gains obtained by combining the streams show that they preserve complementary identity-discriminative cues.

As shown in Table 10, modality labels are most readily decoded from Branch0. Relative to Branch0, the balanced accuracy of Branch1 and Branch2 decreases by 4.58–7.57 percentage points across the two settings. Together with their comparable retrieval performance, this result indicates that the parallel fusion streams reduce linear modality decodability without a marked loss of identity discrimination. Nevertheless, the above-chance AUC values show that modality-related information remains measurable in every stream.

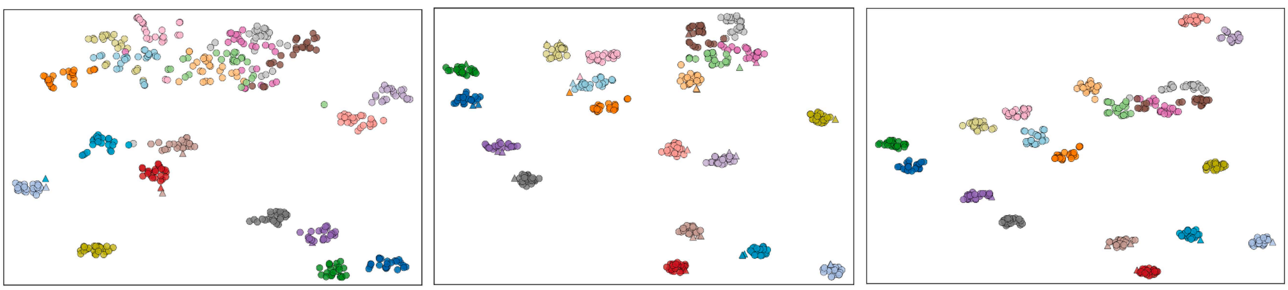
Table 10
Identity-disjoint modality probing on SYSU-MM01 using the seed-0 model. Values are mean±SD over five identity-disjoint folds with balanced visible and infrared samples. Lower scores indicate weaker linear modality decodability.

Feature	All Search		Indoor Search	
	Balanced Acc. (%)	AUC (%)	Balanced Acc. (%)	AUC (%)
Branch0	93.96±3.02	98.83±1.33	91.02±4.33	97.46±2.07
Branch1	88.23±1.96	95.81±1.45	83.45±6.81	**91.73±2.29**
Branch2	88.33±3.26	95.68±1.74	86.44±7.13	94.43±3.12
BN_Sum	**87.90±1.50**	**95.56±1.29**	**82.95±5.90**	92.02±1.70

Table 11 reports cross-modal identity margins and linear HSIC, where HSIC is used only as a proxy for measured linear modality dependence. Branch0 has the highest same-identity similarity and the largest identity margin under both search settings, showing the strongest cross-modal identity separation among the three streams. Branch1 and Branch2 retain positive identity margins while exhibiting lower HSIC values: Branch1 gives the lowest HSIC under All Search, whereas Branch2 gives the lowest value under Indoor Search. For trial 0, the identity-stratified permutation test rejects the null hypothesis of independence between each stream and the modality labels ($p = 0.005$), showing that measurable modality dependence remains in all three streams. This test evaluates each stream separately and does not compare the statistical differences between streams. Together with the retrieval and modality-probe results, these statistics show that the three streams differ in their balance between identity separation and mea-

Table 11
Cross-modal proxy statistics for MBUE streams on SYSU-MM01 using the seed-0 model. Values are mean±SD over 10 gallery trials. S^+ and S^- denote same- and different-identity cross-modal cosine similarities, respectively, with $\Delta = S^+ - S^-$. Lower HSIC indicates weaker linear modality dependence.

Setting	Feature stream	S^+	S^-	Δ	HSIC ($\times 10^{-3}$)
All Search	Branch0	0.6756±0.0055	0.0819±0.0012	0.5937±0.0059	5.607±0.339
	Branch1	0.6426±0.0046	0.0785±0.0009	0.5640±0.0052	5.291±0.302
	Branch2	0.6423±0.0046	0.0782±0.0009	0.5641±0.0051	5.585±0.326
Indoor Search	Branch0	0.6415±0.0050	0.0723±0.0017	0.5692±0.0045	9.071±0.353
	Branch1	0.6035±0.0044	0.0709±0.0016	0.5325±0.0039	8.717±0.453
	Branch2	0.6066±0.0041	0.0720±0.0016	0.5345±0.0037	8.273±0.467



(a) Branch0 (b) Branch1 (c) Branch2

Fig. 4. t-SNE visualization of Branch0, Branch1, and Branch2 on SYSU-MM01 under the all search setting using the seed-0 model. The same identities and samples are used in all panels. Different colors denote identities; triangles denote visible samples and circles denote infrared samples.

sured modality dependence, supporting functional differentiation and partial functional decoupling in MBUE.

As t-SNE (Rauber et al., 2016) provides a two-dimensional projection, Fig. 4 is used only for a qualitative comparison of the three feature streams. For the displayed samples, Branch1 and Branch2 form more compact identity clusters than Branch0, while visible and infrared samples from several identities remain close in the projected space. Branch2 also shows clearer inter-class separation than Branch1, although cross-modal gaps and isolated samples are still visible. These patterns complement the retrieval, modality-probe, and proxy-statistics results by illustrating that the parallel fusion streams retain identity structure while exhibiting different degrees of measured modality dependence.

Across the four analyses, Branch1 and Branch2 exhibit lower linear modality decodability and weaker measured modality dependence than Branch0, while retaining comparable retrieval performance and positive cross-modal identity margins. The t-SNE visualization provides a qualitative view of the corresponding representation structures. This combined evidence shows that MBUE learns complementary representations with different functional tendencies and promotes partial functional decoupling between identity-related and modality-related information.

4.6. Efficiency analysis

We compare the practical efficiency of DIGCA with the DEEN (Zhang & Wang, 2023) baseline and HOS-Net (Qiu et al., 2024) in terms of parameter count, FLOPs, inference latency, and peak GPU memory. All methods are measured on a single NVIDIA RTX A6000 GPU using the same software environment and an input resolution of 384×192. Inference latency is averaged over 100 forward passes with a batch size of 1 after 20 warm-up iterations. CUDA synchronization is performed immediately before and after each timed region. Peak GPU memory is measured using the maximum allocated CUDA memory. The input size, batch configuration, numerical precision, number of data-loading workers, and evaluation interval are kept consistent across the compared methods.

As shown in Table 12, compared with DEEN, DIGCA introduces 37.17M additional parameters and 18.61G additional FLOPs. Its inference latency increases from 5.54 to 11.36 ms, while peak GPU memory increases from 29.72 to 32.86 GB, reflecting the additional computational cost of complementary feature organization and guided cross-stream integration. Compared with HOS-Net, DIGCA uses 6.28M fewer parameters and reduces inference latency by 1.98 ms, but requires more FLOPs and GPU memory. Overall, DIGCA incurs higher computational cost than DEEN. Compared with HOS-Net, it has fewer parameters and lower inference latency, but higher FLOPs and GPU-memory consumption. Together with the preceding retrieval results, these measurements reflect the trade-off between retrieval performance and computational cost.

4.7. Visualization

Feature distributions. Fig. 5(a–c) compares the intra-class and inter-class distance distributions of the initial features, the DEEN baseline, and DIGCA. Here, δ_1, δ_2, and δ_3 denote the differences between the corresponding mean inter-class and intra-class distances. Compared with the initial features and DEEN, DIGCA produces less overlap between the two distributions and increases the distance gap to $\delta_3 = 0.4280$. This result shows that DIGCA improves the separation between same-identity and different-identity cross-modal samples. The t-SNE projections (Rauber et al., 2016) in Fig. 5(d–f) qualitatively show the corresponding feature distributions. In the initial distribution, visible and infrared samples are separated primarily by modality, while samples from different identities overlap substantially. DEEN produces more discernible identity clusters, but visible and infrared samples of several identities remain separated into modality-specific subclusters. With DIGCA, samples belonging to the same identity become more compact across modalities, while samples from different identities are more clearly separated. Together with the distance distributions, these observations indicate that DIGCA improves identity discriminability and reduces the separation between cross-modal samples of the same identity.

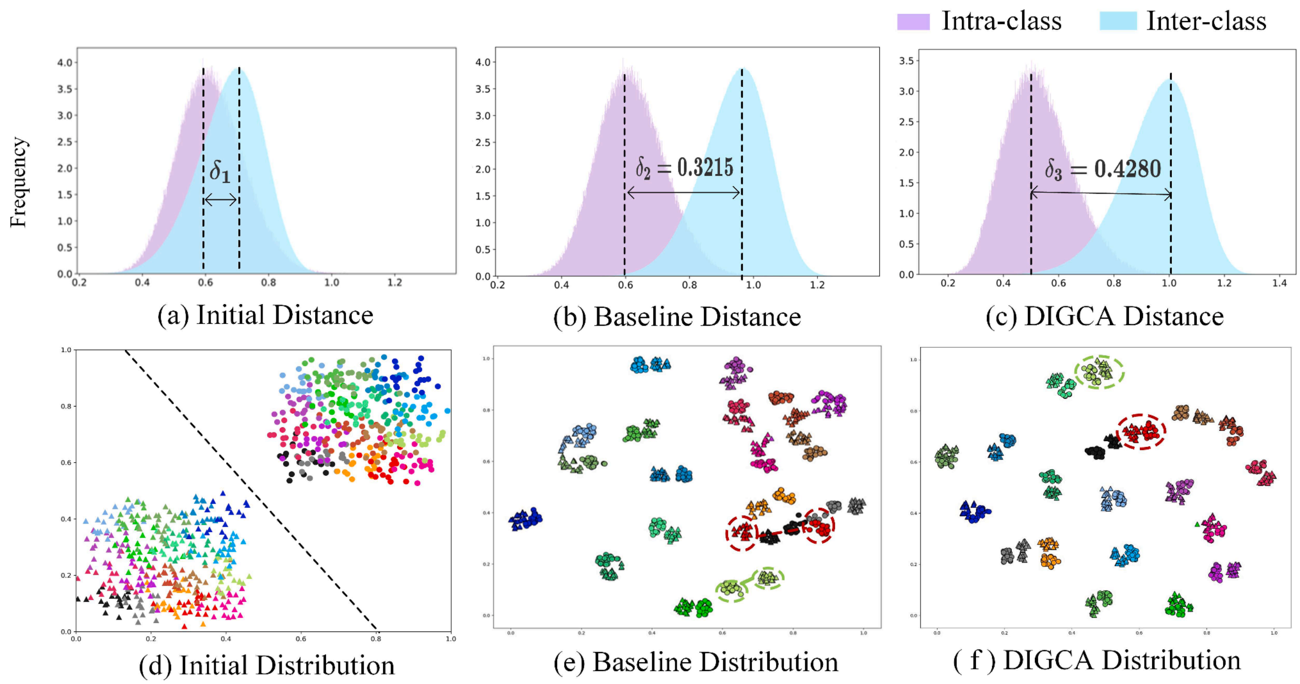


Fig. 5. (a-c) illustrate the changes in intra-class and inter-class distances of cross-modal features. Intra-class and inter-class distances are represented by light purple and light blue, respectively. (d-f) visualize the feature distributions in a two-dimensional space using the t-SNE method, where different colors correspond to different identities, and ○ and △ denote pedestrian images from the VIS and IR modalities, respectively.

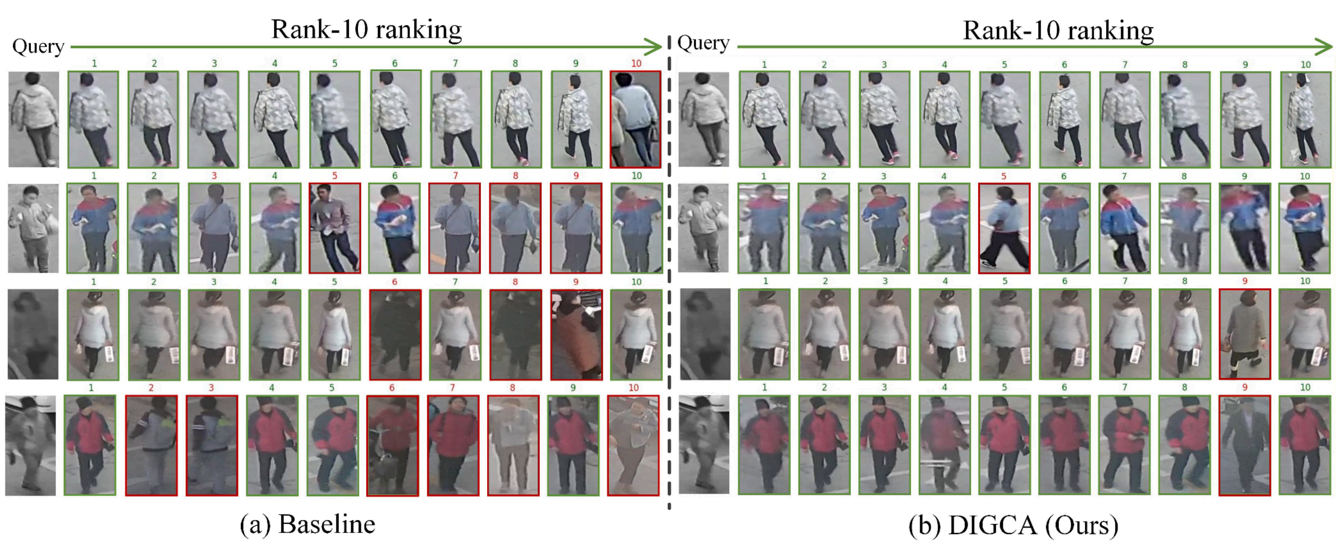


Fig. 6. Partial Rank-10 retrieval results of baseline and DIGCA in the VIS to IR mode on the LLCM dataset.

Retrieval results. Fig. 6(a–b) compares the Rank-10 retrieval results of DEEN and DIGCA under the VIS-to-IR setting on LLCM. For the displayed queries, DIGCA produces fewer incorrect matches than DEEN, and the correct matches generally appear at higher ranks. This improvement is particularly evident for queries affected by low illumination and substantial cross-modal appearance differences. The results indicate that DIGCA provides more discriminative retrieval features and improves the robustness of cross-modal identity matching under challenging low-light conditions.

4.8. Limitation and future work discussion

Despite its competitive results, DIGCA has two limitations. First, it depends on reliable cross-modal identity annotations, which are difficult to obtain from asynchronous or non-overlapping cameras. As shown in Fig. 7, the identities of the retrievals marked in orange cannot be confirmed from the available LLCM annotations. Incomplete labels may also cause unlabeled positive pairs to be treated as negatives, introducing supervision noise and affecting the interpretation of retrieval results. Future work will investigate uncertainty-aware pseudo-labeling, robust objectives, and temporal and camera metadata to improve learning under incomplete supervision.

Second, the complementary feature organization and guided cross-stream integration introduce additional computational cost relative to DEEN, which may limit real-time or resource-constrained deployment. The evaluation is also restricted to three benchmarks and does not fully cover broader variations in camera networks, illumination, and acquisition conditions. Future work will explore parameter shar-

Table 12
Efficiency analysis of DIGCA on the SYSU-MM01 dataset. Latency is measured with batch size 1 during inference, while GPU memory is measured under the same training setting.

Configuration	Params (M) ↓	FLOPs (G) ↓	Latency (ms) ↓	Peak GPU Mem. (GB) ↓
DEEN (Zhang & Wang, 2023)	**89.04**	18.47	**5.54**	29.72
HOS-Net (Qiu et al., 2024)	132.49	**9.16**	13.34	**28.97**
DIGCA (Ours)	126.21	37.08	11.36	32.86

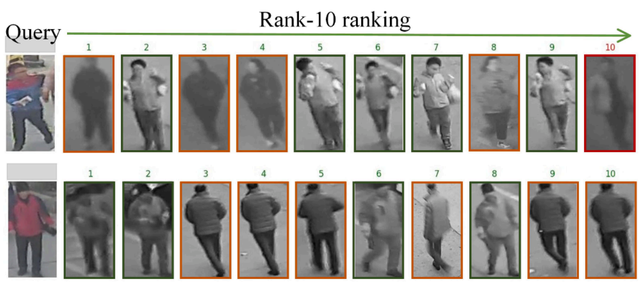


Fig. 7. Partial Rank-10 retrieval results of DIGCA on the LLCM dataset under the IR-to-VIS mode.

ing, lightweight attention, adaptive stream selection, and knowledge distillation, together with evaluation under more diverse cross-modal conditions.

5. Conclusion

This paper presents the Decoupled Information-Guided Cross-Modal Alignment (DIGCA) framework. Its central idea is to enrich the initial identity representation through multi-scale contextual modeling, organize complementary features with different information tendencies, and then use these features to guide cross-stream integration for cross-modal representation alignment. Specifically, HCE aggregates multi-scale contextual information from different receptive fields; MBUE organizes global-relation and local-spatial cues into complementary feature streams and promotes partial functional decoupling; and DIG-CMA refines and integrates these streams through channel attention, spatial attention, and cross-attention. Experiments on SYSU-MM01, RegDB, and LLCM show that DIGCA achieves competitive overall performance compared with recent VI-ReID methods.

CRediT authorship contribution statement

Qingsong Deng: Writing – original draft, Methodology, Conceptualization; **Guangqian Kong:** Writing – review & editing, Validation, Software; **Xun Duan:** Data curation; **Huiyun Long:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

Alehdaghi, M., Josi, A., Cruz, R. M. O., Shamsolmoali, P., & Granger, E. (2025). Adaptive generation of privileged intermediate information for visible-infrared person re-identification. *IEEE Transactions on Information Forensics and Security*, *20*, 3400–3413.
Chan, S., Meng, W., Bai, C., Hu, J., & Chen, S. (2024). Diverse-feature collaborative progressive learning for visible-infrared person re-identification. *IEEE Transactions on Industrial Informatics*, *20*, 7754–7763.
Chen, Y., Wan, L., Li, Z., Jing, Q., & Sun, Z. (2021). Neural feature search for rgb-infrared person re-identification. In *Cvpr* (pp. 587–597).
Dai, W., Lu, L., & Li, Z. (2025). Diffusion-based synthetic data generation for visible-infrared person re-identification. In *Aaai* (pp. 11185–11193).
Deng, J., Dong, W., Socher, R., Li, L. J., Li, K., & Fei-Fei, L. (2009). Imagenet: A large-scale hierarchical image database. In *Cvpr* (pp. 248–255).
Du, B., Liu, H., Wang, G., Guo, J., Lei, Z., & Li, S. Z. (2018). RegDB: A benchmark dataset for RGB-infrared person re-identification. In *Icip* (pp. 3774–3778).
Fan, M., Lai, S., Huang, J., Wei, X., Hu, X., & Ling, H. (2021). Rethinking bisenet for real-time semantic segmentation. In *Cvpr* (pp. 9716–9725).
Feng, J., Wu, A., & Zheng, W. S. (2023a). Shape-erased feature learning for visible-infrared person re-identification. In *Cvpr* (pp. 22752–22761).
Feng, Y., Chen, F., Sun, G., Li, J., & Wang, Z. (2025a). Learning multi-granularity representation with transformer for visible-infrared person re-identification. *Pattern Recognition*, *164*, 111510.
Feng, Y., Chen, F., Yu, J., Ji, Y., Wu, F., Liu, S., & Jing, X. Y. (2025b). Homogeneous and heterogeneous relational graph for visible-infrared person re-identification. *Pattern Recognition*, *158*, 110981.
Feng, Y., Li, J., Yan, Y., Lu, Y., & Zhang, X. (2023b). Visible-infrared person re-identification via cross-modality interaction transformer. *IEEE Transactions on Multimedia*, *25*, 7647–7659.
Guo, J., Du, H., Hao, X., Wang, Y., & Li, S. (2025a). CFET: A cross-fusion enhanced transformer for visible-infrared person re-identification. *Expert Systems with Applications*, *271*, 126645.
Guo, Z., Bian, L., Wei, H., Li, J., Ni, H., & Huang, X. (2025b). DSnet: A novel way to use atrous convolutions in semantic segmentation. *IEEE Transactions on Circuits and Systems for Video Technology*, *35*, 3679–3692.
Hao, X., Zhao, S., Ye, M., & Shen, J. (2021). Cross-modality person re-identification via modality confusion and center aggregation. In *Iccv* (pp. 16383–16392).
He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *CVPR* (pp. 770–778).
Hermans, A., Beyer, L., & Leibe, B. (2017). In defense of the triplet loss for person re-identification, arXiv:1703.07737.
Hua, X., Cheng, K., Lu, H., Wang, Y., & Li, S. (2025). MSCMNet: Multi-scale semantic correlation mining for visible-infrared person re-identification. *Pattern Recognition*, *159*, 111090.
Kim, M., Kim, S., Park, J., Park, S., & Sohn, K. (2023). Partmix: Regularization strategy to learn part discovery for visible-infrared person re-identification. In *Cvpr* (pp. 18621–18632).
Liang, T., Jin, Y., Liu, W., & Li, Y. (2023). Cross-modality transformer with modality mining for visible-infrared person re-identification. *IEEE Transactions on Multimedia*, *25*, 8432–8444.
Liu, H., Xia, D., & Jiang, W. (2023b). Towards homogeneous modality learning and multi-granularity information exploration for visible-infrared person re-identification. *IEEE Journal of Selected Topics in Signal Processing*, *17*, 545–559.
Liu, J., Sun, Y., Zhu, F., Pei, H., Yang, Y., & Li, W. (2022b). Learning memory-augmented unidirectional metrics for cross-modality person re-identification. In *Cvpr* (pp. 19344–19353).
Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., & Guo, B. (2021). Swin transformer: Hierarchical vision transformer using shifted windows. In *Iccv* (pp. 9992–10002).
Lu, H., Zou, X., & Zhang, P. (2023). Learning progressive modality-shared transformers for effective visible-infrared person re-identification. In *AAAI* (pp. 1835–1843).
Lu, Z., Lin, R., & Hu, H. (2024). Tri-level modality-information disentanglement for visible-infrared person re-identification. *IEEE Transactions on Multimedia*, *26*, 2700–2714.
Lu, Z., Lin, R., & Hu, H. (2025). Disentangling modality and posture factors: Memory-attention and orthogonal decomposition for visible-infrared person re-identification. *IEEE Transactions on Neural Networks and Learning Systems*, *36*, 5494–5508.
Qiu, L., Chen, S., Yan, Y., Xue, J. H., Wang, D. H., & Zhu, S. (2024). High-order structure based middle-feature learning for visible-infrared person re-identification. In *Aaai* (pp. 4596–4604).
Rauber, P. E., Falcão, A. X., & Telea, A. C. (2016). Visualizing time-dependent data using dynamic t-sne. In *Eurovis* (pp. 73–77).
Ren, K., & Zhang, L. (2024). Implicit discriminative knowledge learning for visible-infrared person re-identification. In *Cvpr* (pp. 393–402).

Sun, R., Chen, L., Zhang, L., Xie, R., & Gao, J. (2024). Robust visible-infrared person re-identification based on polymorphic mask and wavelet graph convolutional network. *IEEE Transactions on Information Forensics and Security*, *19*, 2800–2813.

Tao, M., Long, H., Kong, G., & Duan, X. (2025). Combining information augmentation aggregation and dual-granularity feature fusion for visible-infrared person re-identification. *Signal, Image and Video Processing*, *19*, 163.

Wang, G., Zhang, T., Cheng, J., Liu, S., Yang, Y., & Hou, Z. (2019a). RGB-infrared cross-modality person re-identification via joint pixel and feature alignment. In *Iccv* (pp. 3623–3632).

Wang, G., Zhang, T., Yang, J., Cheng, J., Chang, J. l., Liang, X., & Hou, Z. (2020). Cross-modality paired-images generation for rgb-infrared person re-identification. In *Aaai* (pp. 12144–12151).

Wang, Z., Wang, Z., Zheng, Y., Chuang, Y. Y., & Satoh, S. (2019b). Learning to reduce dual-level discrepancy for infrared-visible person re-identification. In *Cvpr* (pp. 618–626).

Wei, Z., Yang, X., & Wang, N. (2024). Dual-adversarial representation disentanglement for visible infrared person re-identification. *IEEE Transactions on Information Forensics and Security*, *19*, 2186–2200.

Wei, Z., Yang, X., Wang, N., Dong, J., Gong, Y., & Li, H. (2021). Syncretic modality collaborative learning for visible infrared person re-identification. In *Iccv* (pp. 225–234).

Wu, A., Zheng, W. S., Yu, H. X., Gong, S., & Lai, J. (2017). RGB-infrared cross-modality person re-identification. In *Iccv* (pp. 5390–5399).

Wu, J., Liu, H., Shi, W., Liu, M., & Li, W. (2024). Style-agnostic representation learning for visible-infrared person re-identification. *IEEE Transactions on Multimedia*, *26*, 2263–2275.

Wu, Q., Dai, P., Chen, J., Lin, C. W., Wu, Y., Huang, F., Zhong, B., & Ji, R. (2021). Discover cross-modality nuances for visible-infrared person re-identification. In *Cvpr* (pp. 4330–4339).

Yang, M., Liu, J., Sun, Y., Zhu, F., Pei, H., & Yang, Y. (2022). Learning with twin noisy labels for visible-infrared person re-identification. In *Cvpr* (pp. 14308–14317).

Yang, X., Dong, W., Li, M., Wei, Z., & Wang, N. (2024). Cooperative separation of modality shared-specific features for visible-infrared person re-identification. *IEEE Transactions on Multimedia*, *26*, 8172–8183.

Yang, Z., Wu, X., & Yang, Y. (2025). Multi-dimensional feature learning for visible-infrared person re-identification. *Big Data Research*, *40*, 100522.

Ye, M., Ruan, W., Du, B., Lin, G., & Xiang, T. (2021a). Channel augmented joint learning for visible-infrared recognition. In *Iccv* (pp. 13567–13576).

Ye, M., Shen, J., Lin, G., Xiang, T., Shao, L., & Hoi, S. C. (2021). Deep learning for person re-identification: A survey and outlook. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *44*, 2872–2893.

Ye, M., Wu, Z., Chen, C., & Du, B. (2024). Channel augmentation for visible-infrared re-identification. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *46*, 2299–2315.

Yu, J., Feng, Y., You, S., Jing, Z., Wu, F., Ji, Y., & Jing, X. Y. (2026). Occlusion-aware cross-modality completion network for occluded visible-infrared person re-identification. *IEEE Transactions on Circuits and Systems for Video Technology*. in press. https://doi.org/10.1109/TCSVT.2026.3688708

Yu, M., Ge, Y., Chen, Z., You, R., Zhu, L., Lin, M., & Xu, Z. (2025). No escape: Towards suggestive clues guidance for cross-modality person re-identification. *Information Fusion*, *122*, 103185.

Zhang, Q., Lai, C., Liu, J., Wang, X., & Lin, H. (2022). FMCNet: Feature-level modality compensation for visible-infrared person re-identification. In *Cvpr* (pp. 7349–7358).

Zhang, Y., & Wang, H. (2023). Diverse embedding expansion network and low-light cross-modality benchmark for visible-infrared person re-identification. In *Cvpr* (pp. 2153–2162).

Zhang, Y., Yan, Y., Li, J., Lu, Y., & Zhang, X. (2023). MRCN: A novel modality restitution and compensation network for visible-infrared person re-identification. In *Aaai* (pp. 3498–3506).

Zhang, Y., Yan, Y., & Lu, Y. (2021). Towards a unified middle modality learning for visible-infrared person re-identification. In *Mm* (pp. 788–796).

Zhang, Y., Yan, Y., Lu, Y., Li, J., & Zhang, X. (2025). Adaptive middle modality alignment learning for visible-infrared person re-identification. *International Journal of Computer Vision*, *133*, 2176–2196.

Zuo, X., Peng, J., Cheng, T., & Wang, H. (2025). Modality-perceptive harmonization network for visible-infrared person re-identification. *Information Fusion*, *118*, 102979.